WORKED SOLUTION

STEP I 2006, Question 14

Probability and Statistics • Sampling Without Replacement

STEP I 2006, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s1-q14

The key idea

Taking the red sweet on the rth attempt means $r-1$ blues followed by the red. With replacement every draw looks the same, so the probability is a simple product of powers; we then maximise over n by treating n as a continuous variable and differentiating the logarithm. Without replacement the cleanest view is to imagine drawing *all* the sweets: the red one is then equally likely to be in any position.

Solution

Throughout, $n \geq 0$ is an integer, and R is the number of the attempt on which the red sweet is first taken.

Part (i)

Because the sweet is returned and the bag shaken, each draw is independent of the others, and on every draw the red sweet is taken with probability $\frac{1}{n+1}$ and a blue one with probability $\frac{n}{n+1}$. The event $R = r$ means the first $r-1$ draws are blue and the rth is red, so by independence

$$P(R=r) = \left(\frac{n}{n+1}\right)^{r-1} \frac{1}{n+1} = \frac{n^{r-1}}{(n+1)^r}, \qquad r = 1, 2, 3, \ldots$$

(This is the geometric distribution with success probability $\frac{1}{n+1}$.)

Now fix r and think of this as a function of n:

$$f(n) = \frac{n^{r-1}}{(n+1)^r}.$$

If $r = 1$ then $f(n) = \frac{1}{n+1}$, which is largest at $n = 0$ (with no blue sweets, the first sweet is certainly red). So suppose $r \geq 2$. Then $f(0) = 0$ and $f(n) > 0$ for $n > 0$, so we may

take logarithms on $n > 0$, treating n as a real variable:

$$\ln f(n) = (r-1)\ln n - r\ln(n+1), \qquad \frac{\mathrm{d}}{\mathrm{d}n}\ln f(n) = \frac{r-1}{n} - \frac{r}{n+1} = \frac{(r-1)-n}{n(n+1)}.$$

Ask yourself n has to be a whole number. Is it legitimate to differentiate?

Yes, because we use the derivative only to find where f increases and decreases. For $n > 0$ the denominator $n(n+1)$ is positive, so the derivative is positive for $0 < n < r-1$ and negative for $n > r-1$. Hence f is strictly increasing on $[0, r-1]$ and strictly decreasing on $[r-1, \infty)$. Since $r-1$ is itself an integer, it beats every other integer, and there is no need to compare neighbouring integers.

So in every case (including $r = 1$) the probability is maximised by

$$n = r - 1,$$

with maximum value $\dfrac{(r-1)^{r-1}}{r^r}$ (read as 1 when $r = 1$). For example, with $r = 3$ the best bag has $n = 2$ blue sweets and gives probability $\frac{4}{27}$.

Part (ii)

Now the draws are not independent, but there are two easy routes.

Direct product. For $R = r$ we need blue on each of the first $r-1$ draws and then red. This is only possible if there are enough blues, $r - 1 \le n$. Given that the earlier draws were all blue, the kth draw is made from a bag of $n + 2 - k$ sweets, one of them red. So

$$P(R = r) = \frac{n}{n+1} \cdot \frac{n-1}{n} \cdot \frac{n-2}{n-1} \cdots \frac{n-r+2}{n-r+3} \cdot \frac{1}{n-r+2}.$$

Each numerator cancels the next denominator, and the product telescopes to

$$P(R = r) = \begin{cases} \dfrac{1}{n+1}, & 1 \le r \le n+1, \\ 0, & r > n+1. \end{cases}$$

Symmetry. Stopping at the red sweet makes no difference to which position it occupies. So imagine carrying on until the bag is empty. All $(n+1)!$ orders of the (distinguishable) sweets are equally likely, and so the red sweet is equally likely to be in any of the $n+1$ positions. That gives the same answer.

For the maximisation, fix r. If $n < r-1$ the probability is 0. For $n \ge r-1$ it is $\frac{1}{n+1}$, which strictly decreases as n increases. So the probability is maximised by the smallest admissible bag,

$$n = r - 1,$$

and the maximum probability is $\frac{1}{r}$. (With $n = r-1$ the red sweet is equally likely to be any of the r sweets, and drawing it last is one of the r equally likely outcomes.)

Checks

- In (i), $\sum_{r\geq 1} P(R=r) = \frac{1}{n+1}\cdot\dfrac{1}{1-n/(n+1)} = 1$, and in (ii) the $n+1$ equal values $\frac{1}{n+1}$ sum to 1.
- With $r=3$ in (i): $n=1,2,3$ give $\frac{1}{8}$, $\frac{4}{27}\approx 0.148$ and $\frac{9}{64}\approx 0.141$, so $n=2$ wins.
- We checked (ii) by listing every ordering of the sweets for $n\leq 6$. A simulation of (i) with $n=3$, $r=2$ gave 0.188, against the exact $\frac{3}{16}=0.1875$.

Common slips

- Writing $\left(\frac{n}{n+1}\right)^r \frac{1}{n+1}$ in (i): only $r-1$ blues come before the red.
- In (ii), forgetting that the answer is 0 when $r > n+1$. This is exactly what stops "make n as small as possible" from running away to $n=0$.

Beyond STEP

Both maximisations are really *maximum likelihood estimation*. If you see that the red sweet first appears on attempt r but don't know n, the value of n that makes your observation most probable is $n=r-1$ in both experiments. This is a cornerstone idea of first-year statistics. The symmetry argument in (ii) is an example of *exchangeability*, which makes many sampling-without-replacement problems easy. Both ideas are developed in Blitzstein and Hwang, *Introduction to Probability*.

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